ELEC5305- Removing Unwanted Noise with Beamforming

Student Information:

Full Name: John Dul
Student ID: jdul9044
Github Username: Jdul9044
Github Project Link: https://github.com/Jdul9044/ELEC5305-Project-Jdul9044

Project Overview

Currently in Digital Signal Processing (DSP), particularly in audio and acoustics the quality of a signal still remains to be an area of study and for future development. One way in understanding quality is to investigate its Signal to Noise Ratio (SNR). An audio recording with a high level of noise can contaminate the audibility and potentially make adjusting the gain redundant. In an amplifier if there is a low SNR in the input of the system as the gain is increased, the noise is also increased, hence making it hard to remove noise when it is already in the system. For higher grade audio equipment, a lower noise floor is more attractive for studio application, however, for inexpensive products this may not be achievable. There are multitude techniques applied into optimising and improving SNR. One way is to "add some spatial information to the time frequency" which is achieved using multiple sensors, this is called beamforming [1]. For the basis of this project the intention is to demonstrate how a conventional beamformer can achieve SNR improvements in a noisy environment and possibly evolving the computation into an adaptive beamformer such as Minimum Variance Distortionless Response (MVDR).

Background and Motivation

The use of beamforming was a technique not well understood until the early 1900s by Guglielmo Marconi and Karl Ferdinand Braun and then later in the 1960s became introduced in sonar phased array systems [2]. After the 1960s adaptive beamforming methods were starting to be implemented which essentially are algorithms that adjust based on the environment through statistics and attempt to negate other interference or noise [2]. Although beamforming has mostly been a solved problem, standard conventional beamformers are still used today for their robustness. In addition, there also has been studies made on improving robustness as well on MVDR [2]. Due to the advancements of faster Analogue to Digital Converters (ADC) and easier to obtain electronics, access to digital beamforming has made both types of beamforming widely used today [3]. This DSP technique was chosen for exploration of complexity and for its wide use in audio application.

Literature Review

Beamforming, although a very powerful technique in improving a SNR of signal, can come with limitations especially for a linear array configuration. One limitation is mentioned in "Beamforming Using Digital Piezoelectric Mems Microphone Array" which details how spatial aliasing can be caused by geometric constraints [4]. Spatial aliasing can be described as a phenomenon where directional information is limited to a particular frequency band and can cause artifacting in the data from grating lobes.

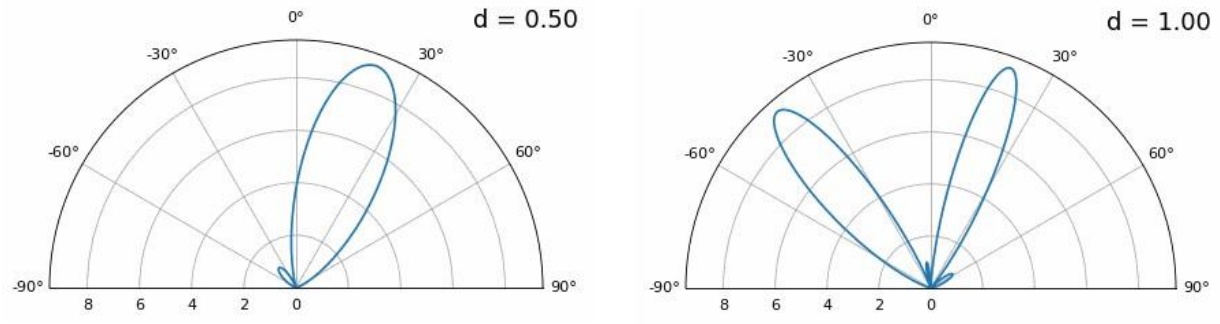


Figure 1. (left) Beamforming without spatial aliasing, (right) grating lobes becoming introduced [5]

Consequently, spatial aliasing can have a profound effect on the data if not well understood. To beamform with an optimum performance and avoid spatial aliasing the paper suggests keeping an array with at least a spacing of less than half a wavelength as shown by the following [6]:

$$d \leq \frac{\lambda}{2}$$

Where d is the element spacing and λ corresponds to the wavelength. As a result, the best theoretical spacing to avoid grating lobes can be found by the following:

$$d \leq \frac{\left(\frac{c}{f_{max}}\right)}{2}$$

Conversely, the beamformer performance on the lower frequency end can be reduced, this is due to the fundamental limitation of angular resolution. The angular resolution determines how the main lobe beamwidth is adjusted. The paper "Simulation and design of a microphone array for beamforming on a moving acoustic source" quotes how the main lobe is widest" at the minimum frequency [6]. Hence, for a narrower main lobe the distance spacing should also not be too low.

Additionally, the paper titled "Angular Resolution Enhancement Provided by Nonuniformly-Spaced Linear Hydrophone Arrays in Ocean Waveguide Remote Sensing" explores how the use of multiple arrays with different spacings into one can achieve performance improvements in the angular resolution [7]. By performing beamforming with different element spacing the main lobe widening can be prevented.

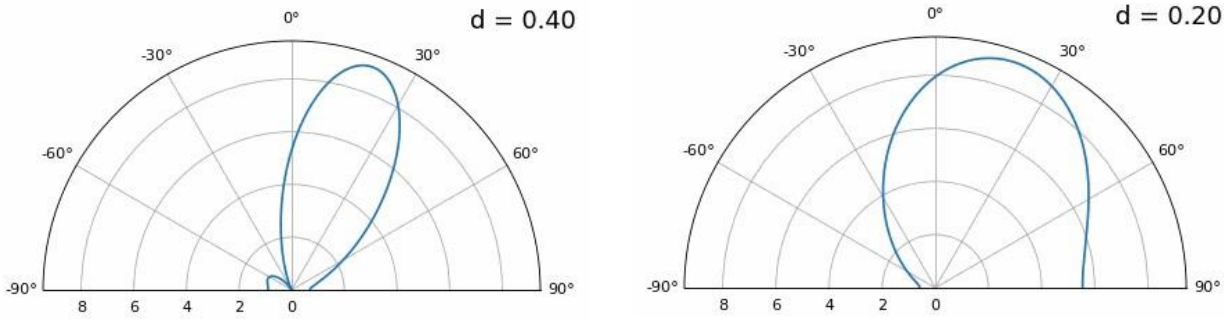


Figure 2. The effect of spacing on angular resolution [5]

Conventional beamformers tend to have the most robust implementation, however, utilise uniform weighting that can potentially include sidelobes that could affect the data. The use of spatial tapering can be applied for the number of elements. Although side lobe reduction is improved, there is a trade-off with the main lobe beamwidth being increased and the gain being reduced [7]. In addition, the paper "Hierarchical Beamforming Strategies for Source Detection With Linear Arrays at 26 GHz" talks about how to achieve high sidelobe suppression performance the need for much larger number of elements is required [8].

For a standard linear array one of the problems beamforming can have is inconsistency in the shape of the main lobe across the frequency. For example, "A Three-Dimensional Pattern-Space Representation for Volumetric Arrays" goes in depth in relation to how the main lobe broadens the most at a steering angle facing end-fire [9]. For volumetric and planar array this is not the case as the main lobe remains consistent regardless of the steering angle [9]. One of the main consequences of the linear array is addressed in the paper "Left-Right Ambiguity Resolution of a Towed Array Sonar" which details how a linear array is limited to a "cone" shape restricting the array to 180 degrees, having a "left" or "right" ambiguity [10].

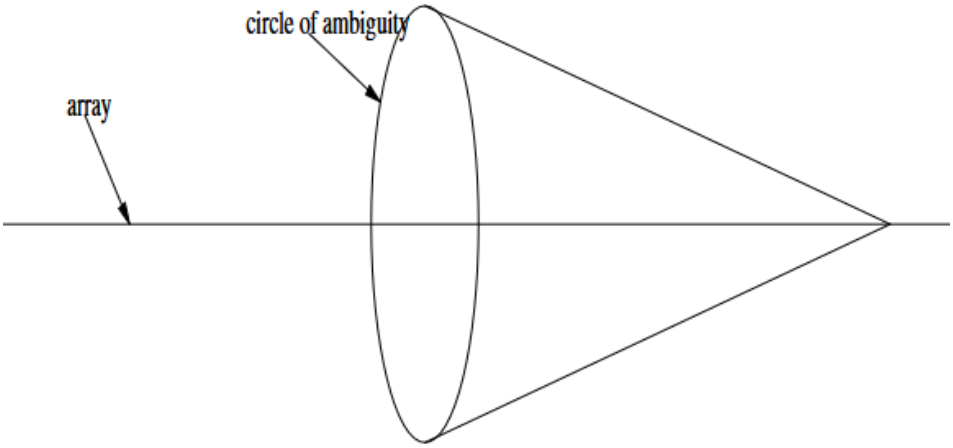


Figure 3. Linear Array Ambiguity [10]

Beamforming with an adaptive beamformer such as MVDR can pose a challenge when faced with errors in measurements such as positional information. "Digital Beamforming with Non-ideal Random Volumetric Arrays" details how the SNR performance of conventional beamformers can potentially be greater than MVDR "in presence of steering vector error" [11]. In addition, having an accurate covariance matrix is another important component to perform MVDR beamforming which is mentioned in "MVDR Algorithm Based on Estimated Diagonal Loading for Beamforming" [12]. This paper addresses on of these fundamental issues by introducing diagonal loading into the covariance matrix to stabilise these problems. Although the use of diagonal matrix can be useful, a small "snapshot" can introduce "large inaccuracy in the sample covariance matrix [12]. Due to the variance of time "snapshots" in the data caused by different number of sensors potentially for different frequency bands, a more adaptable diagonal loading is required [12].

Proposed Methodology

Tools and Platforms

For the main purpose of this project report the analysis will be conducted in Jupyter Labs with Python. Libraries such as Numpy, Scipy and Matplotlib will be used as the main packages. However, generated simulated data will be produced using Python scripts. The main code will be developed for a post-processing environment.

Setup

The current setup of the beamforming array for a simulation will be a fixed spacing for a set of elements. Due to the loss in angular resolution the frequency will also be limited to a low frequency region. For an optimum array for beamforming, the following spacing conditions are required:

$$d \leq \frac{c}{\left(\frac{f_{max}}{2}\right)}$$

$$d \leq \frac{1500}{\left(\frac{400}{2}\right)}$$

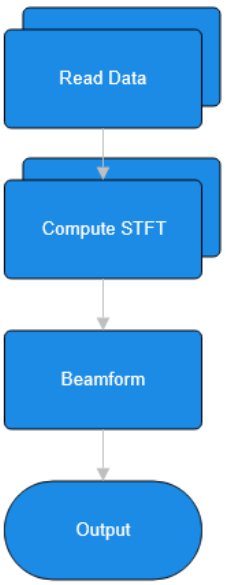
$$d \leq 1.875m$$

The speed of sound in water will be assumed to be at approximately 1500 m/s, with a selected frequency band of 0 to 400 of interest.

For the simplicity of the analysis, a uniform linear array will be tested for the simulation where no angle greater than 180 degrees in azimuth will be selected with a flat 0-degree elevation. Due to the spatial ambiguity constraints of the linear array for real linear arrays especially underwater, the target will not be on the sea floor and be moving above, which will be moving relative to the array axis, thus restricted to an angle of a cone.

DSP Technique

The overall analysis will consist of signal processing chain as followed:



Read Data - The raw data is initially read as an individual audio Pulse Code Modulation (PCM) file for each channel with a bit-depth of at least 16-bit is selected for minimal quantisation effects [13].

Downsample - Due to the enormous size of the samples used for a long duration with a sampling frequency of 48kHz dataset, the data can be downsampled, given the frequency of interest is fixed to the practical geometric limitation of the array for beamforming. By downsampling the data, the post processing time is effectively reduced and optimised.

Compute Short-Time Fourier Transform (STFT) - The analysis will consist of beamforming in the frequency domain is selected as it is the fastest method for computation for post processing. The data that will be used for the beamforming will have STFT coefficients in the form of complex to utilise the phase information and be converted into Power Spectral Density (PSD) for a proper representation of how each bin the energy is stored in a spectrogram. A discrete-time STFT will be the main method of computing the spectrogram with a standard Hann Window and a 50% overlap choice. Window sizes will be selected for narrowband and wideband for performance metrics.

Beamform - A conventional beamformer will be implemented for the frequency domain that will be generalised to work with both linear and volumetric arrays. In addition, the conventional beamformer will be compared for an adaptive beamforming using MVDR.

One of the foundational important components for beamforming needed is finding the time delay with respect to each element to a reference. In 3D space, the projection angle of the planar waves can be determined by finding the dot product of the array geometry with the unit vector which is simply the target of interest in spherical coordinates. This is shown below:

$$\tau = \frac{\vec{r} \cdot \vec{u}}{c}$$

Where $\vec{r}$ is the position matrix of the array in the following form:

$$\vec{r} = \begin{bmatrix} x_1 & y_1 & z_1 \\ \vdots & \vdots & \vdots \\ x_n & y_n & z_n \end{bmatrix}$$

Where $\vec{u}$ is the unit vector following an adjusted spherical coordinate system convention as followed.

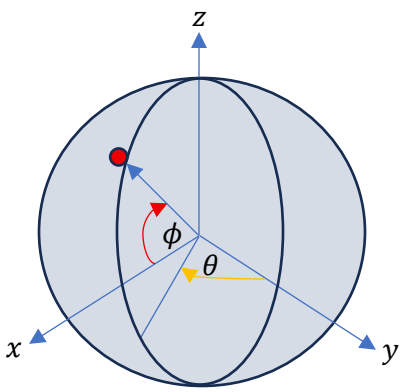


Figure 4. Adjusted spherical coordinate system for underwater [14]

This translates into the following vector:

$$\vec{u} = \begin{bmatrix} \cos\phi \sin\theta \\ \cos\phi \cos\theta \\ \sin\phi \end{bmatrix}$$

Where ϕ is the elevation and θ is the azimuth angles. For the basis of the report, the hydrophone array will be defined by the following convention for reference:



Figure 5. Hydrophone Reference

The hydrophone reference is taken from the centre, where the first half of the array is negative and second half become positive. To compute the steering vector, the phase information is stored as a function of frequency for each element in the vector.

$$s = e^{-j2\pi f \tau}$$

Where s is the steering vector, f is the selected frequency bin and τ being the time delay previously mentioned.

For a conventional beamformer, the weights that are applied to the steering vector. Given the nature of the underwater environments having low SNR, the use of spatial tapering will not be considered for the report. Hence, resulting into beamformer output to the following:

$$Y = conj(s) \cdot X$$

Where the dot product is performed once again to the conjugate of the steering vector with the spectrogram coefficients of each element for a particular time slice.

To implement adaptive beamforming (MVDR) the weights for the steering vector are adapted by the covariance matrix. The "spatial" covariance matrix can be found by the following:

$$R = \frac{XX^H}{N}$$

Where X refers to the spectrogram that is multiplied by itself with the Hermitian matrix divided by N the number of sensor elements. Note that for a non-stationary target, using all the time samples from the recording cannot be used for a single covariance matrix. A new method must be adopted to use multiple time samples which can be done by implementing a moving averaging window. In addition, if the covariance matrix is ill-conditioned as previously discussed, a diagonal loading is applied. For this report, a robust adaptive diagonal loading is applied as followed:

$$w_{mvdr} = \frac{R^{-1}s}{s^H R^{-1} s}$$

$$Y = conj(w_{mvdr}) \cdot X$$

(Optional) Interpolating - For producing the simulation data sample interpolation is required to adjust the time delay on all channels.

Synthetic Production – To test the beamformer, a simulated dataset is produced. To develop the best possible results the simulated dataset will includes tones as well as a broadband inject signals into the files. The pure tone can be produced by the following:

$$x(t) = \text{Asin}(2\pi f(t-\tau)$$

For a broadband signal:

$$x(t) = FFT^{-1}\left(X(f,t)e^{-1j2\pi f\tau}\right)$$

Where $X(f,t)$ is a spectrogram that contains random noise generated between a frequency band.

Results

Before testing real data, a simulation of multiple targets is performed to test adaptive beamforming capability. The simulation consisted of two closely spaced signals where target one was a single 250 Hz generated tone at 10 degrees azimuth whereas the 2nd target had broadband with also a single tone at 300 Hz at 60 degrees azimuth. Figure 5 shows the azimuth angle corresponding to each target whilst assuming elevation is 0 degrees.

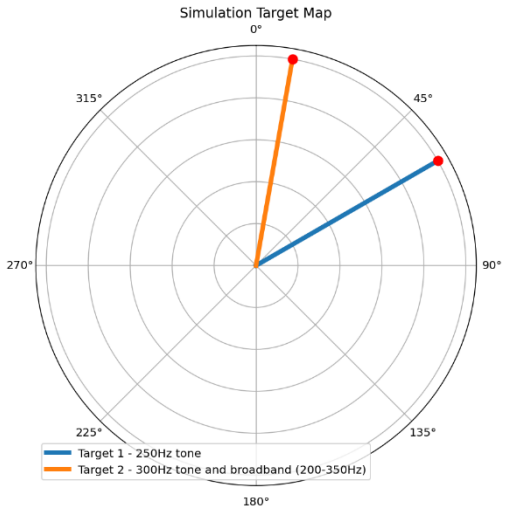


Figure 6. Simulation Target Map

From the two targets a standard single hydrophone cannot distinguish the directivity alone which can be seen in figure 6 where the SNR is lowest among all methods. By utilising 8 elements of the artificially generated hydrophones, the beamformer can suppress the background, hence increasing the SNR as illustrated here. It can be seen that since the beam steers on target one, not only the background is suppressed but also both beamforming methods attempt to reduce interference (target two) with MVDR having superior results in this example.

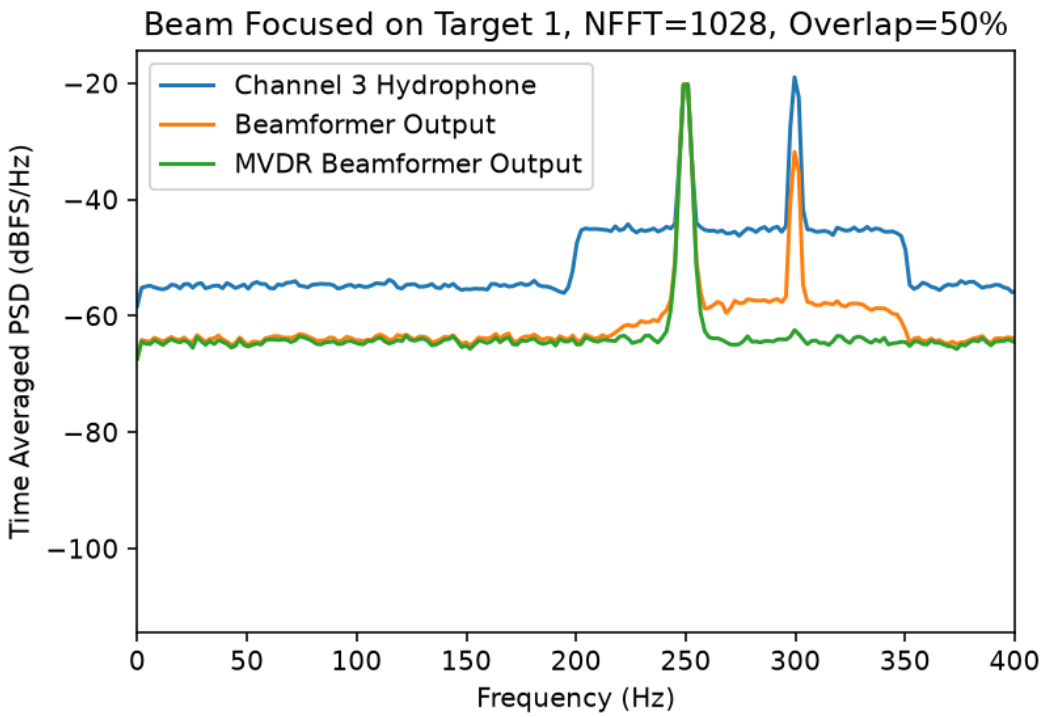


Figure 7. Welch's method graph comparison between all methods for artificial data

The suppression of the interference demonstrates how unwanted noise can be removed using beamformer techniques regardless on the type of signal whether it be a tone or broadband. Figure 7 illustrates a visual representation of a stationary target tone remaining unchanged over time with various techniques mentioned. The beamforming methods show the broadband segment of energy and tones being completely removed in the MVDR display.

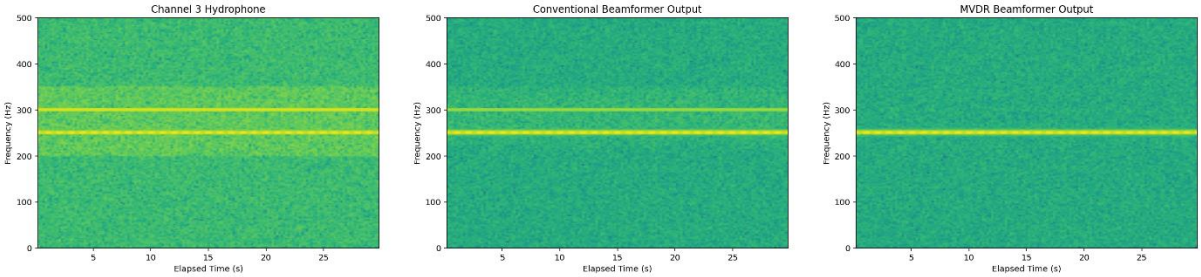


Figure 8. Spectrogram display comparison between all methods for artificial data

To match a real dataset the case of a moving target is considered. A second simulation test is performed by moving target one. The target is moved from its original position at 60 degrees azimuth to 90 degrees as displayed on the simulation moving target in Figure 9.

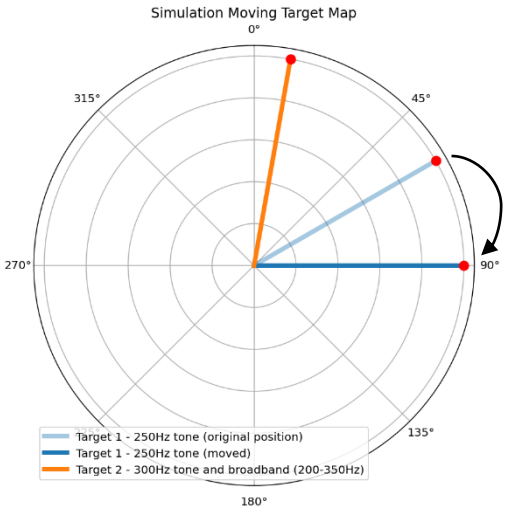


Figure 9. Simulation Moving Target Map

For this analysis a time varying unit vector was required in order keep the beam steered at the target. The results were displayed below.

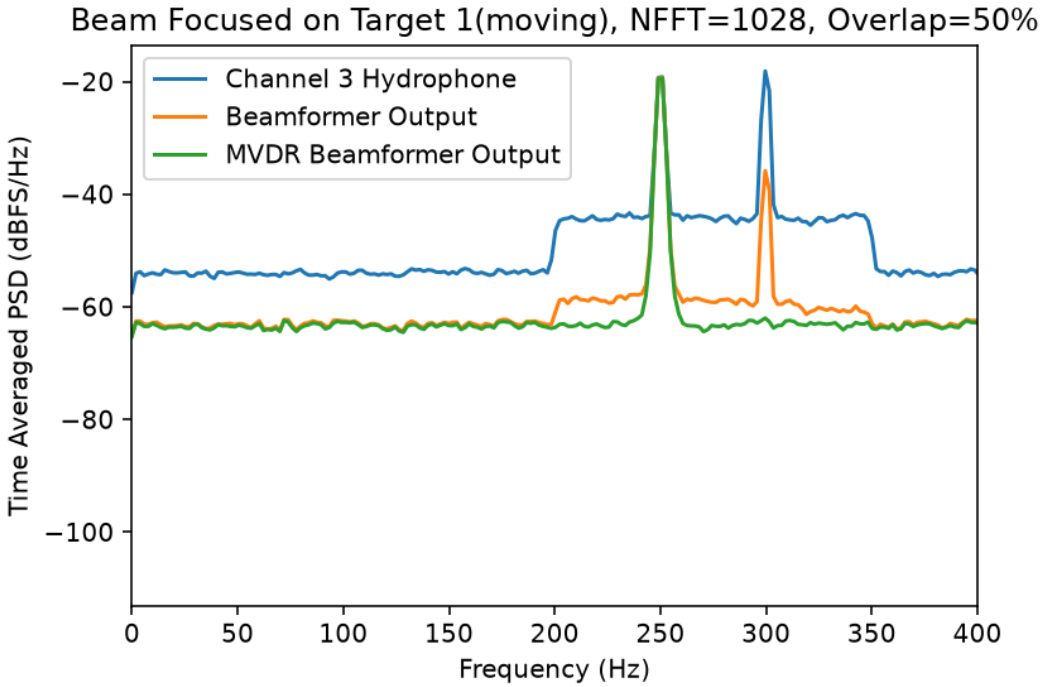


Figure 10. Welch's method graph comparison between all methods for artificial data for moving target

For code validation a package from ARL Python Tools is used to validate and compare both CBF and MVDR beamformer produced results using the same synthetic dataset. For the snapshots required for the covariance matrix in MVDR, the simulation tested various window length to compare performance as well as adjusting the adaptive loading to regularise the matrix.

	Mean Tolerance (dB)	Window Length	Adaptive Loading
CBF	1.47e-15	N/A	N/A
MVDR	0.77	81	1
MVDR	0.91	81	0.1
MVDR	0.92	81	1e-3

MVDR	1.47	21	0.1
MVDR	1.11	41	0.1
MVDR	0.97	61	0.1

Table 1. Validation Results

For one selection of time snapshots for a frequency bin the quality of inverse covariance matrix can be compared.

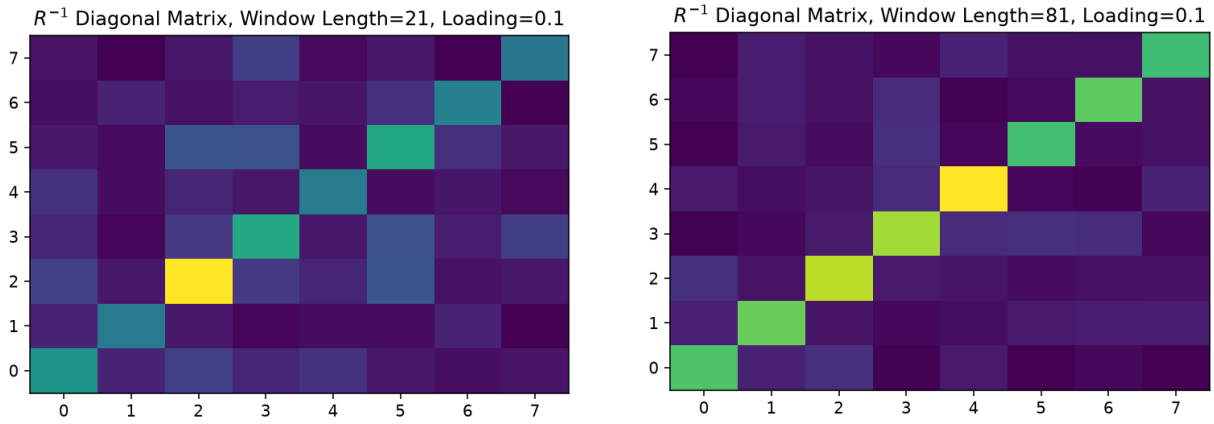


Figure 11. Inverse Covariance Matrix Comparison

The real dataset was analysed using SWellEx-96 experiment which comprised of 4 arrays, one of interest was the Horizontal Linear Array (HLA) for north of the exercise for event S59. The experiment consisted of 32 elements, with 27 working and an aperture of 240m long which was deployed from the north to south with the y-axis mainly used [15]. However, due to the deployment of the array, the "horizontal linear array" had variation in the x-axis and potentially can be considered a random linear array with a fixed depth of 213 meters [15]. A map can be seen in figure 12 which highlights the arrays and includes a target noise source and interfere [15]. A MATLAB script is run to read the "sio" file format where all 27 hydrophones are selected with 100,000 samples. This translates into approximately 30 seconds recording. In addition the acquisition system of the HLA North is listed as recording at 3276.8 Hz [15].

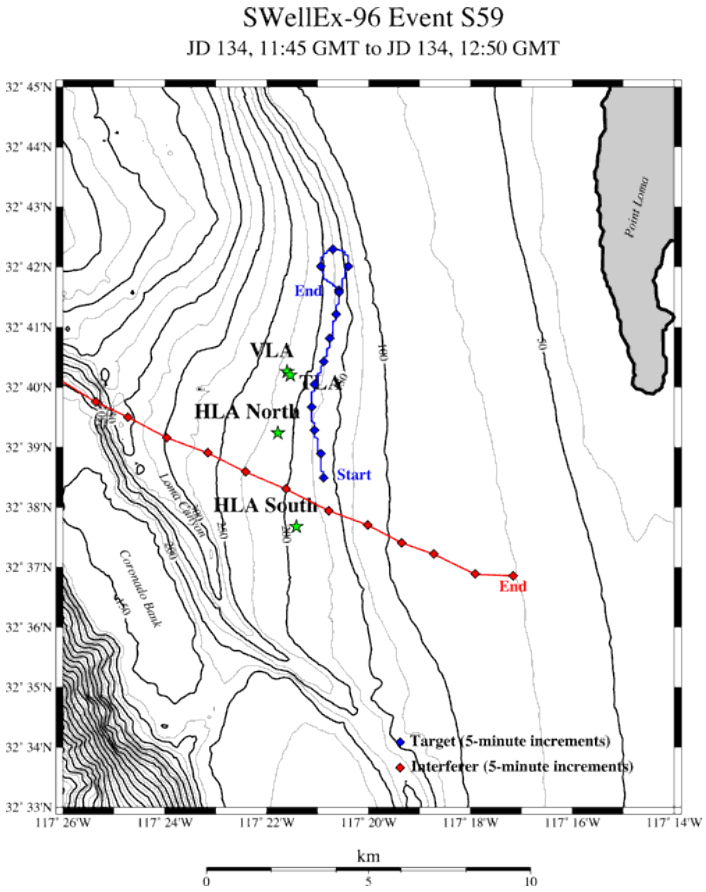


Figure 12. Exercise S59 Map Overview [15]

For the basis of the report….

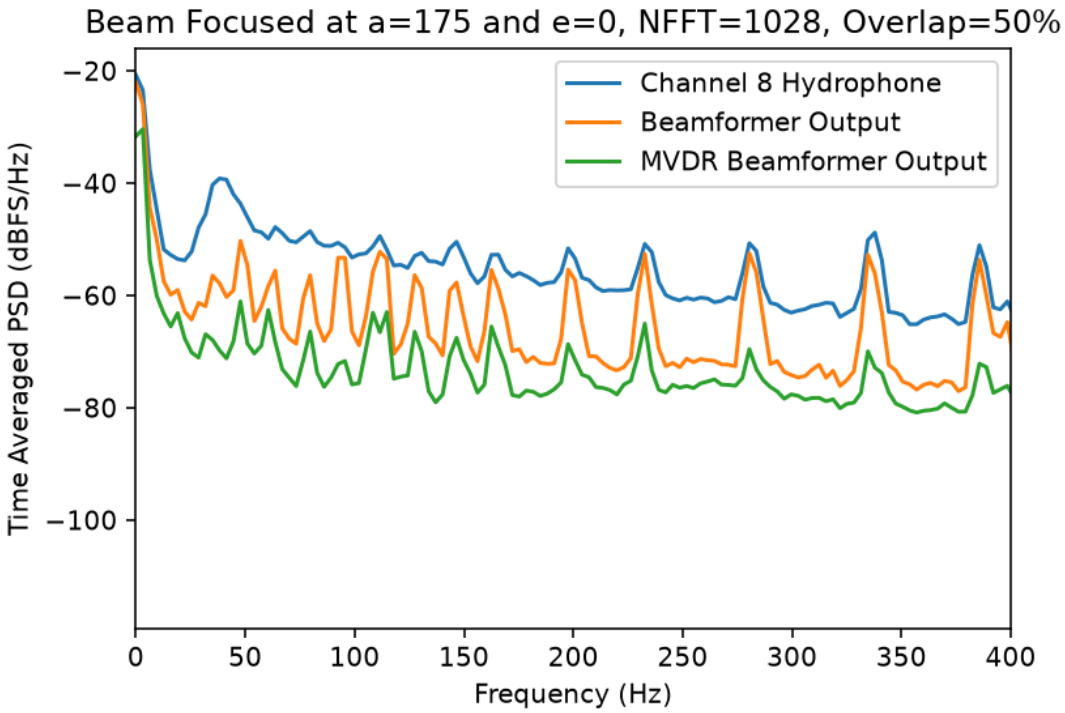


Figure 13. Real dataset comparison

Mention lowest spacing(add here)

Mention validation(add here)

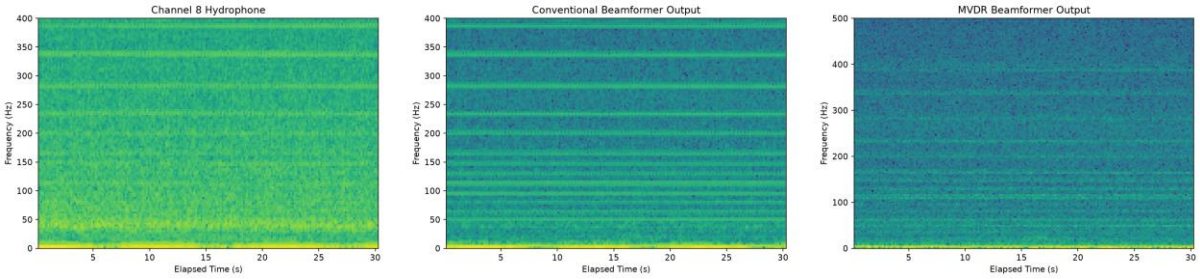


Figure 14. Spectrogram comparison for real dataset

Discussion

To be done.

Conclusion

To be done.

Reference

[1] A. Hidri, S. Meddeb, A. Alaqeeli, and H. Amiri, “Beamforming techniques for multichannel audio signal separation,” *J. Digit. Commun. Technol.*, 2012. [Online]. Available: https://arxiv.org/abs/1212.6080

[2] A. M. Elbir, K. V. Mishra, S. A. Vorobyov, and R. W. Heath, “Twenty-five years of advances in beamforming: From convex and nonconvex optimization to learning techniques,” *IEEE Signal Process. Mag.*, vol. 40, no. 4, pp. 118–131, Jun. 2023, doi: 10.1109/MSP.2023.3262366.

[3] S. K. Sharma *et al.*, “Cognitive beamforming for spectral coexistence of hybrid satellite systems,” in *Cooperative and Cognitive Satellite Systems*, 2015, ch. 13.

[4] [Online]. Available: https://api.research-repository.uwa.edu.au/ws/portalfiles/portal/265325932/2111.10087v1.pdf

[5] PySDR, “Beamforming and direction of arrival (DOA).” [Online]. Available: https://pysdr.org/content/doa

[6] [Online]. Available: https://www.acoustics.asn.au/conference_proceedings/AAS2013/papers/p30.pdf

[7] [Online]. Available: https://www.mdpi.com/2072-4292/9/10/1036

[8] [Online]. Available: https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2024RS008215

[9] [Online]. Available: https://commons.erau.edu/cgi/viewcontent.cgi?article=1103&context=publication

[10] [Online]. Available: https://ora.ox.ac.uk/objects/uuid:68cae061-75f7-4619-a7f3-e9b7003e04f4/files/m15cb6d0b21eacfb01639ea91d2a7f49e

[11] [Online]. Available: https://etda.libraries.psu.edu/files/final_submissions/28?remediated=true

[12] [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1155/2017/7904356

[13] R. Triggs, “Audio bit depth: Everything you need to know,” *SoundGuys*, Dec. 17, 2024. [Online]. Available: https://www.soundguys.com/audio-bit-degree-explained-23706/

[14] *Engineering Statics*, “Coordinates in 3D.” [Online]. Available: https://engineeringstatics.org/coordinates-3d.html

[15] Scripps Institution of Oceanography, “SWELLEX-96: North horizontal line array.” [Online]. Available: https://swellex96.ucsd.edu/hla_north.htm